

PSET 8: Multivariate distributions — Solutions

Problem 1: Properties of a joint PDF

$f_{XY}(x, y) = kx^3y^2$ for $0 < x, y < 6$.

a. Find k .

Ans.: Since f_{XY} factors into a function of x times a function of y over a rectangular support, split the double integral:

$$k(\int_0^6 x^3 dx)(\int_0^6 y^2 dy) = 1$$

$$\int_0^6 x^3 dx = [x^4/4]_0^6 = 1296/4 = 324$$

$$\int_0^6 y^2 dy = [y^3/3]_0^6 = 216/3 = 72$$

$$k(324)(72) = 1 \Rightarrow k(23328) = 1$$

$$k = 1/23328$$

b. Find the marginal PDF of X , $f_X(x)$.

$$\text{Ans.}: f_X(x) = \int_0^6 kx^3y^2 dy = kx^3 \int_0^6 y^2 dy = kx^3(72) = (72/23328)x^3$$

$$f_X(x) = x^3/324, \text{ for } 0 < x < 6$$

c. Find the marginal PDF of Y , $f_Y(y)$.

$$\text{Ans.}: f_Y(y) = \int_0^6 kx^3y^2 dx = ky^2 \int_0^6 x^3 dx = ky^2(324) = (324/23328)y^2$$

$$f_Y(y) = y^2/72, \text{ for } 0 < y < 6$$

(Note: since $f_{XY}(x,y) = f_X(x)f_Y(y)$ here, X and Y are actually independent in this problem.)

d. Find $\Pr(X < 2, Y > 4)$.

$$\text{Ans.}: \Pr(X < 2, Y > 4) = \int_0^2 \int_4^6 kx^3y^2 dy dx = k(\int_0^2 x^3 dx)(\int_4^6 y^2 dy)$$

$$\int_0^2 x^3 dx = [x^4/4]_0^2 = 16/4 = 4$$

$$\int_4^6 y^2 dy = [y^3/3]_4^6 = (216-64)/3 = 152/3$$

$$\Pr(X < 2, Y > 4) = (1/23328)(4)(152/3) = 608/69984 = 19/2187 \approx 0.00869$$

Problem 2: Calculating conditional PDF

$f(x, y) = 18x^2y^2$ for $0 \leq x \leq y \leq 1$.

a. Sketch the region on which $f(x, y)$ is positive.

Ans.: The support is not the full unit square — it is the triangular region above the line $y = x$: bounded by $x = 0$ (left edge), $y = 1$ (top edge), and the diagonal $y = x$ (lower-right boundary). For any fixed x , y ranges only from x up to 1, not from 0 to 1.

b. Find $f(x|y)$.

Ans.: First find the marginal $f_Y(y)$. For fixed y , x ranges from 0 to y :

$$f_Y(y) = \int_0^y 18x^2y^2 dx = 18y^2[x^3/3]_0^y = 18y^2(y^3/3) = 6y^5, \text{ for } 0 < y < 1$$

$$f(x|y) = f(x,y)/f_Y(y) = 18x^2y^2 / 6y^5 = 3x^2/y^3$$

$$f(x|y) = 3x^2/y^3, \text{ for } 0 \leq x \leq y$$

c. Are X and Y independent? Explain your reasoning, paying attention to the region on which $f(x, y)$ is defined.

Ans.: No. Two things confirm this:

- The support itself is not rectangular. Independence requires the joint support to be a product of the individual supports. Here the constraint $x \leq y$ ties the two variables together — the possible values of X depend on the value of Y (e.g. if $Y = 0.2$, X must lie in $[0, 0.2]$, but if $Y = 0.9$, X can range up to 0.9).
- Algebraically, $f(x|y) = 3x^2/y^3$ depends on y. If X and Y were independent, $f(x|y)$ would equal $f_X(x)$ alone with no y in the expression.

Even when a joint PDF factors into an x-part times a y-part, that alone is not sufficient for independence — the support must also factor into a rectangle. Here it does not, so X and Y are dependent.

Problem 3: Moments and independence

$f_{WZ}(w, z) = kw^2z^3$ for $0 < w, z < 2$.

a. Find k.

Ans.: $k(\int_0^2 w^2 dw)(\int_0^2 z^3 dz) = 1$

$$\int_0^2 w^2 dw = [w^3/3]_0^2 = 8/3$$

$$\int_0^2 z^3 dz = [z^4/4]_0^2 = 4$$

$$k(8/3)(4) = 1 \Rightarrow k(32/3) = 1$$

$$k = 3/32$$

b. Find $E[W]$.

Ans.: $f_W(w) = \int_0^2 kw^2z^3 dz = kw^2(4) = (3/32)(4)w^2 = (3/8)w^2$, for $0 < w < 2$

$$E[W] = \int_0^2 w(3/8)w^2 dw = (3/8)\int_0^2 w^3 dw = (3/8)(4) = 3/2$$

c. Find $\text{Var}(W)$.

Ans.: $E[W^2] = \int_0^2 w^2(3/8)w^2 dw = (3/8)\int_0^2 w^4 dw = (3/8)(32/5) = 12/5$

$$\text{Var}(W) = E[W^2] - (E[W])^2 = 12/5 - (3/2)^2 = 2.4 - 2.25 = 0.15 = 3/20$$

d. Find $\text{Cov}(W, Z)$.

Ans.: $f_Z(z) = \int_0^2 kw^2z^3 dw = kz^3(8/3) = (3/32)(8/3)z^3 = (1/4)z^3$, for $0 < z < 2$

$$E[Z] = \int_0^2 z(1/4)z^3 dz = (1/4)\int_0^2 z^4 dz = (1/4)(32/5) = 8/5 = 1.6$$

$$E[WZ] = \int_0^2 \int_0^2 wz \cdot kw^2z^3 dw dz = k(\int_0^2 w^3 dw)(\int_0^2 z^4 dz) = (3/32)(4)(32/5) = 12/5 = 2.4$$

$$\text{Cov}(W, Z) = E[WZ] - E[W]E[Z] = 2.4 - (1.5)(1.6) = 2.4 - 2.4 = 0$$

e. Are W and Z independent? Explain your reasoning using mathematical concepts from the course.

Ans.: Yes. The support $0 < w < 2, 0 < z < 2$ is a rectangle — the range of W does not depend on Z and vice versa (unlike Problem 2). The joint PDF also factors cleanly: $f(w, z) = (3/8)w^2 \times (1/4)z^3$, exactly the product of the marginals $f_W(w)$ and $f_Z(z)$, which is the definition of independence.

($\text{Cov}(W, Z) = 0$ is consistent with independence but is not sufficient evidence on its own, since zero covariance only rules out linear dependence — the support and factorization argument is what establishes full independence here.)

f. What is the PDF of W conditional on Z, $f_{W|Z}(w|z)$?

Ans.: $f_{W|Z}(w|z) = f(w,z)/f_Z(z) = kw^2z^3 / [(1/4)z^3] = kw^2(4) = (3/8)w^2$

$f_{W|Z}(w|z) = (3/8)w^2$, for $0 < w < 2$

This equals $f_W(w)$ exactly, with no z in it — a direct confirmation of independence.

Problem 4: Properties of joint random variables

$E[D] = 8$, $E[F] = 4$, $E[DF] = 10$, $\text{Var}(D) = 30$, $\text{Var}(F) = 60$.

a. What is $\text{Cov}(D, F)$?

Ans.: $\text{Cov}(D,F) = E[DF] - E[D]E[F] = 10 - (8)(4) = 10 - 32 = -22$

b. What is the correlation between D and F?

Ans.: $\text{Cor}(D,F) = \text{Cov}(D,F) / \sqrt{(\text{Var}(D)\text{Var}(F))} = -22 / \sqrt{(30 \cdot 60)} = -22 / \sqrt{1800} \approx -22/42.43 \approx -0.519$

c. What is $\text{Var}(D + F)$?

Ans.: $\text{Var}(D+F) = \text{Var}(D) + \text{Var}(F) + 2\text{Cov}(D,F) = 30 + 60 + 2(-22) = 90 - 44 = 46$

d. What is $\text{Var}(D - F)$?

Ans.: $\text{Var}(D-F) = \text{Var}(D) + \text{Var}(F) - 2\text{Cov}(D,F) = 90 - (-44) = 90 + 44 = 134$

e. Suppose you multiplied F by 2 to generate a new variable, H. What is $\text{Cov}(D, H)$?

Ans.: $H = 2F$, so $\text{Cov}(D,H) = \text{Cov}(D, 2F) = 2\text{Cov}(D,F) = 2(-22) = -44$

f. What is $\text{Cor}(D, H)$? How does this compare to your answer to part (b) of this question?

Ans.: $\text{sd}(H) = 2 \cdot \text{sd}(F)$, so:

$\text{Cor}(D,H) = \text{Cov}(D,H) / (\text{sd}(D) \cdot \text{sd}(H)) = -44 / (\text{sd}(D) \cdot 2\text{sd}(F)) = \text{Cov}(D,F) / (\text{sd}(D) \cdot \text{sd}(F)) = \text{Cor}(D,F) \approx -0.519$

This is identical to the answer in part (b). Correlation is invariant to positive linear rescaling of either variable, since the scaling factor cancels between the covariance and the standard deviation.

g. Suppose instead that $\text{Var}(D) = 40$. How would this change $\text{Cor}(D, F)$?

Ans.: $\text{Cor}(D,F) = -22 / \sqrt{(40 \cdot 60)} = -22 / \sqrt{2400} \approx -22/48.99 \approx -0.449$

The magnitude of the correlation decreases (moves closer to 0) relative to part (b), since $\text{Cov}(D,F)$ is unchanged but the denominator $\sqrt{(\text{Var}(D)\text{Var}(F))}$ increases.

Problem 5: Continuous Bayes' theorem

Building on Problem 2, where $f(x|y) = 3x^2/y^3$ (for $0 \leq x \leq y$) and $f_Y(y) = 6y^5$ (for $0 < y < 1$).

a. Using $f(x|y)$ and $f_Y(y)$, apply Bayes' theorem to find $f(y|x)$. Be careful with the limits of integration in the denominator.

Ans.: $f(y|x) = f(x|y)f(y) / \int f(x|y)f(y) dy$

Numerator: $f(x|y)f(y) = (3x^2/y^3)(6y^5) = 18x^2y^2$

Denominator — careful with limits: since x is fixed and known, y ranges from x up to 1 (not 0 to 1), because the support requires $x \leq y \leq 1$:

$$\int_x^1 18x^2y^2 dy = 18x^2[y^3/3]_x^1 = 18x^2(1-x^3)/3 = 6x^2(1-x^3)$$

$$f(y|x) = 18x^2y^2 / [6x^2(1-x^3)] = 3y^2/(1-x^3)$$

$$f(y|x) = 3y^2/(1-x^3), \text{ for } x \leq y \leq 1$$

b. Compute $f(y|x)$ directly as $f(x,y)/f_X(x)$ and confirm it matches part (a). What familiar quantity does the denominator of Bayes' theorem turn out to be here?

Ans.: First find $f_X(x)$. Since y ranges from x to 1 for fixed x :

$$f_X(x) = \int_x^1 18x^2y^2 dy = 18x^2[y^3/3]_x^1 = 6x^2(1-x^3), \text{ for } 0 < x < 1$$

This is exactly the denominator computed in part (a) — not a coincidence.

$$f(y|x) = f(x,y)/f_X(x) = 18x^2y^2 / [6x^2(1-x^3)] = 3y^2/(1-x^3)$$

This matches part (a) exactly.

The denominator of Bayes' theorem, $\int f(X|\theta)f(\theta) d\theta$, is nothing other than $f_X(x)$ — the marginal density of the data (the normalizing constant ensuring $f(y|x)$ integrates to 1). This mirrors the discrete case, where the denominator of Bayes' theorem is the total probability $P(B) = \sum P(B|A)P(A)$ — here it is the continuous analogue, obtained by integrating out θ (i.e. y) instead of summing over discrete events.